FANUC P-50*i*B R-30*i*B Mate Plus

Startup Guide

Thank you for purchasing a FANUC Robot

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

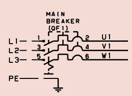
- Straight-head screwdriver
- Cross-head screwdriver
- Torque wrench
- Metric hex-head wrenches

Required Items

- Properly rated transport equipment as specified
- Properly rated power source as specified on the Controller Power Source Label
- Rigid platform to hold robot
- Properly rated input power cable Conductor size of AWG14 to AWG10 terminal size is M5
- 4-M20 x 35mm bolts
- 4-M20 flat washers
- Multimeter
- Loctite #243

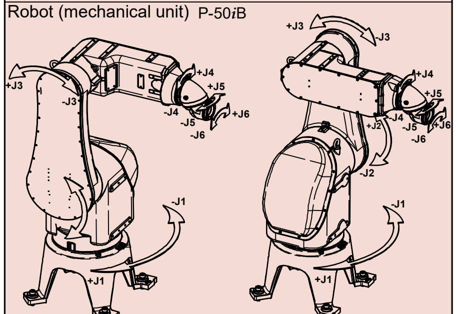
Three Phase

380-575 VAC



R-30*i*B Mate Plus Controller

Robot (mechanical unit) P-50*i*B



Right Arm Offset

Left Arm Offset

A. Unpack/Install your Robot

Controller Weight: 83kg

Robot Weight: 323kg

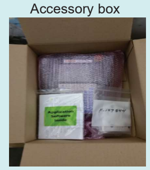
Shipping Contents

1. Unbox the pallet, cut any shipping straps securing the robot, and unwrap the robot.

2. Open the large box on the pallet. In the large box you will find an accessory box, Power Cables, Ground Cables and a Purge Air Hose.

Cable & Hose boxes

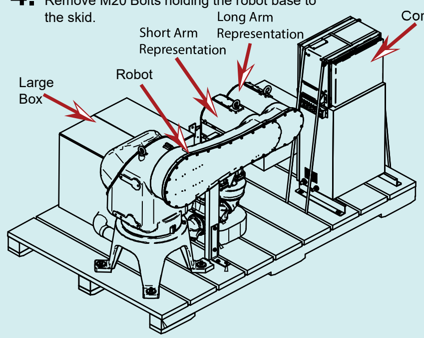
Accessory box



3. In the accessory box you will find a hardware bag, Software, eDocs, Teach Pendant, & Controller drawing.

Keep this information in a safe, known location.

4. Remove M20 Bolts holding the robot base to the skid.

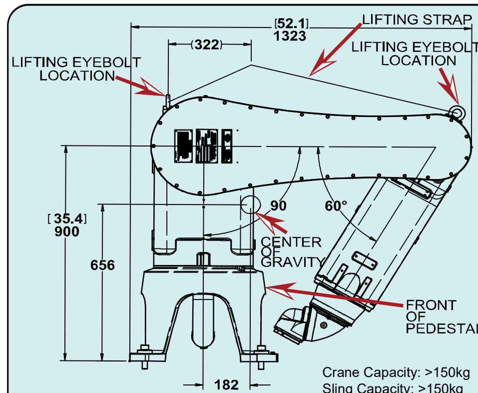


⚠ WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

5. Use properly rated equipment to move the robot to its new location.

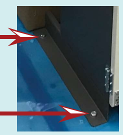
Note: If using a sling (strap), make sure to use the Eye Bolts. Make sure that the lifting strap is secure and taut while lifting.



Crane Capacity: >150kg

Sling Capacity: >150kg

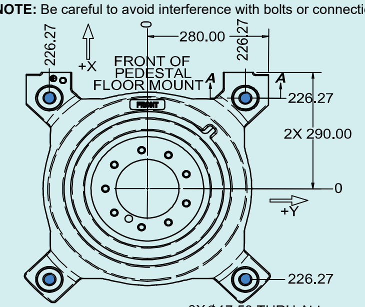
6. Remove the bolts securing the Controller to the transportation equipment (shown below).

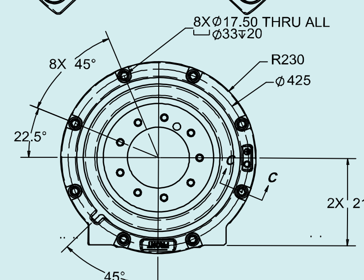


7. Use 4-M20x50 bolts with washers to bolt the robot to a rigid platform or table that will prevent any motion of the robot base. Secure the robot using the M20 Bolts. Tighten to 546 N-m (402 ft-lbs of torque)

See diagram below for the P-50*i*B mounting pattern. More information is available in the P-50*i*B Mechanical Unit Operator's Manual.

NOTE: Be careful to avoid interference with bolts or connections





Robot installation is complete.
You are now ready to go to **B. Unpack/Install your Controller**

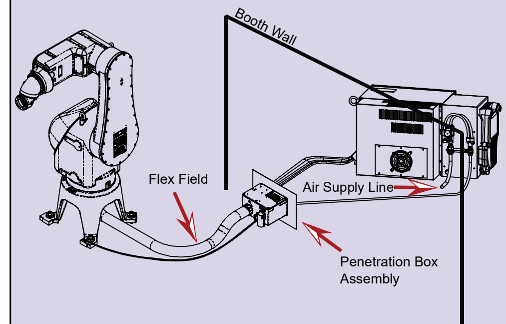
B. Unpack and Install your Controller

1. Use properly rated equipment to move the controller to its new location.

⚠ WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

Penetration Box Assembly Diagram



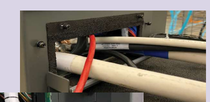
2. Find the power source needed for your controller on the door near the F Number. See diagram to the left for F Number location. Connect the controller to power.

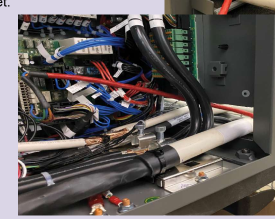
⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

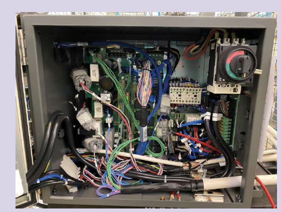
3. Open the controller door by using a straight-head screwdriver to turn the latches clockwise, then turn the Breaker counter-clockwise past its off position.

4. Route the power input cable through the opening on the bottom, right-hand side of the cabinet.





Inside view



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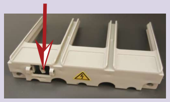
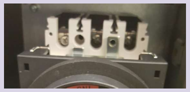
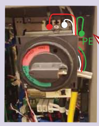
5. To connect power, remove the plastic cover from above the breaker (see diagram below) To remover the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.

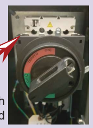

Plastic Breaker cover opening location


Breaker with cover removed

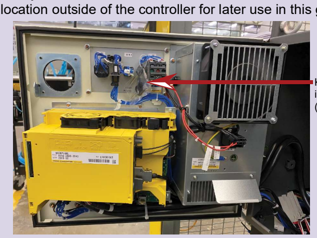
6. Connect power to the Breaker with the customer-supplied power input cable. For three-phase, connect L1 to 1, L2 to 3, and L3 to 5. Connect the ground (connection just to the right of the Breaker). After power is connected, reinsert the plastic cover.

Note: Use a multimeter to verify the correct voltage is present.


Breaker connection with customer-supplied power input cable


Breaker with cover reinserted

7. Before you close the controller door, retrieve the set of keys from inside the door, and store them in a known location outside of the controller for later use in this guide.


Key Location inside controller (on the door)

8. Close the controller door and use a straight-head screwdriver to turn the latch on the outside of the controller counter-clock-wise.

Controller installation is complete. DO NOT TURN ON THE CONTROLLER UNTIL YOU ARE INSTRUCTED TO DO SO. Go to B. Unpack and Install your Robot.

C. Turn on your Controller

⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

1. Rotate the Breaker clockwise to ON.

2. Connect the air supply to the robot. The pressure switch will close and remain closed as long as the air supply maintained.

3. Check that the PURGE FAULT indicator is illuminated. If not, ensure the breaker is on and the controller is receiving power.

4. Press and hold the PURGE ENABLE Button.


Purge Enable Button

Purge Notification Lights

5. The Purge control Module switches the solenoid valve to distribution lines which starts the flow of purge air. Air will be distributed throughout Robot cavity by distribution lines.

6. The flow is monitored by a flow switch at the air outlet. When the flow reaches 700 L/min, the flow switch closes the electronic contacts. Flow Indication will turn on and the control module starts timing. At the same time, the PURGE FAULT indicator turns off.

7. When the Flow Indication turns on, the operator can release the PURGE ENABLE button.

8. Purge air flow continues for 3.5 minutes to ensure air changes. Time is measured by hardware logic in the purge control module.

9. After the purge is complete, the purge control module switches the solenoid valve to maintenance air, closes the output contacts to provide power to the Robot controller, and illuminates the PURGE COMPLETE indicator. The air pressure in the cavity decreases and the check valve closes to prevent loss of maintenance air.

10. Press the POWER ON button to turn on the Controller.

Note: As long as purge is complete, the Robot may be turned on and off as desired.

Robot Purge Air Supply

Minimum Flowswitch Activation: 566 SLPM
Maximum Flowswitch Activation: 740 SLPM
Air Supply Pressure: 60-75 PSI (Dynamic)
Max. Maintenance Flow: 70-175 NLPM
Maintenance Pressure: 1.99-4.23 kPa
Min. Overpressure: 0.172 kPa
Max. Overpressure: 10.3 kPa
Max. Leakage Flow: 175 SLPM
Operating Ambient Temp: 0°C - 45°C
Filtration: 5 Micron Absolute
Exhaust Pressure: 0.149kPa
Air Supply Humidity: -40°C Pressure dewpoint at 90 PSI

Turning on your Controller is complete. You are now ready to go to D. Jog your Robot.

D. Jog your Robot

1. Locate the Mode Switch on the Controller Operator Panel. Use the key retrieved in Step A12 to switch to T1 mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.


Mode Switch

2. Pick up the teach pendant and verify that the controller is in T1 Mode similar to the screen shot below. If it is not, repeat the previous step.

3. Press the COORD key on the teach pendant until JOINT is displayed similar to the screen shot below.

T1 Mode Joint

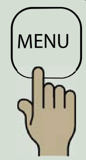
4. To jog the robot in JOINT mode:

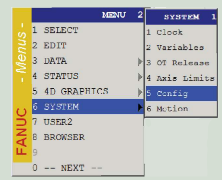
a. Turn the teach pendant ON/OFF switch to the on position, press MENU, move the cursor to ALARM, and press ENTER.

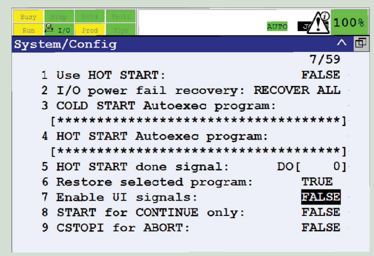
b. To toggle between alarm history and active alarm display, press F3.

c. If the error MCTL-013 ENBL input is off is displayed in the alarm history when you try to move the robot (it will not display as an active alarm), you must disable UI.

To do this:
- Press MENU, Select Next (0), and select (SYSTEM), and then select CONFIG.
- Use the teach pendant arrow keys to move the cursor to the Enable UI signals options.
- Use the teach pendant keys to set the option to FALSE.







d. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.

e. Press and hold the DEADMAN switch on the back of the teach pendant in its middle position. Press RESET on the teach pendant to clear alarms.

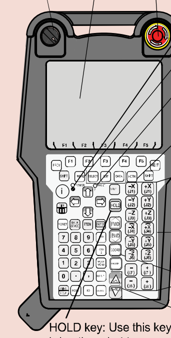
f. To move each joint axis of the robot, Press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the Robot (Mechanical Unit) diagram on page 1 of this guide.

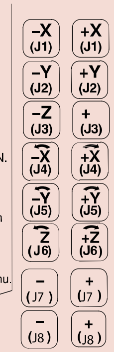
g. Ensure you are able to jog each robot joint.

Turning on and jogging the robot is complete. You are now ready to preform software installation, if necessary, refer to the Controller-specific software installation manual.

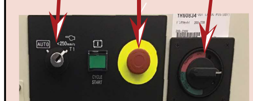
Reference

Your actual teach pendant key locations could vary.



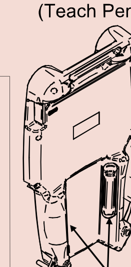


Refer to your Application-Specific Setup and Operations Manual for more information on each teach pendant key.



Controller Operator Panel

*i*Pendant (Teach Pendant)


DEADMAN switch

For more information

To access controller, robot, or application-specific manuals, scan the **QR Code** to the right or go to **https://www.fanucamerica.com/qr/edocs/**.

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